

Cutting MEMS Drive Voltage by 41% by Differentiating Through Pull-In

Tushar Dudgeja, *Student Member, IEEE*, and Nawaz Shafi, *Member, IEEE*
Vellore Institute of Technology, India

Differentiable pull-in solver

solves the saddle-node bifurcation directly · exact gradients, no surrogate fitting · no training data

Inverse design

22.86 → 13.60 V

–41% drive voltage, electrode shape alone

15 s on a laptop CPU

no new mask, no commercial licence

Amortized network

0.04 ms per design

spec → geometry, 0 stored simulations

0.56% spec error

hard-constraint layer, not a penalty

Safe-operation RL

+12.8% usable travel

0.0% of devices destroyed

learns each device's ceiling

which cannot be measured

Problem statement

Electrostatic actuation drives most commercial MEMS: micromirror arrays, RF switches, tunable capacitors and optical attenuators. Two limits bound all of them, and both originate in pull-in.

Drive voltage is the first. Holding a useful deflection safely takes tens of volts, well above any logic supply, so the die carries a charge pump that costs area and quiescent power. Travel is the second: stable motion stops at a third of the gap, at a threshold that moves with film thickness, residual stress and etch bias. That threshold cannot be located without crossing it, and crossing it destroys the device, so the drive schedule is set for the worst die on the wafer and every good die is under-driven.

Designers meet this with FEM sweeps over hand-picked geometry families, in tools that cost a licence and return a number with no derivative. Current AI practice inherits the handicap: the standard pipeline fits a surrogate to 48,000 simulations, then optimises through its approximation error.

The AI approach, and its novelty

We make the physics differentiable, including the instability itself.

Pull-in is a saddle-node bifurcation. At the fold the tangent stiffness is singular, so a differentiated forward solve fails at precisely the point of interest, which is one reason the problem is usually approached with surrogates. Rather than approaching the fold, we solve for it: adding a null-vector condition and a normalisation to the equilibrium residual gives

$$R(Y, \Lambda) = 0, \quad J(Y, \Lambda) v = 0, \quad \|v\| = 1,$$

which turns the bifurcation into an ordinary root of a regular system. Newton lands on it directly, and the implicit function theorem then returns the **exact** derivative of pull-in voltage with respect to every point of the electrode gap, in one reverse pass.

To our knowledge, no existing differentiable solver targets the electrostatic pull-in bifurcation. JAX-FEM, JAX-MPM and JaxSSO return gradients through finite-element, meshfree and structural problems, but none differentiates a fold point. Because the gradient comes from the physics rather than from fitted data, **none of the three capabilities above requires a training set.**

Impact on MEMS

Drive voltage falls 22.86 V → 13.60 V, a 41% reduction, on the same beam meeting the same 2 μm travel specification. The reduction comes entirely from reshaping the electrode gap, so it requires no new material, no additional process step and no additional mask.

The optimiser reaches it in **15 seconds on a laptop CPU** without a commercial licence, starting from a uniform gap and given

no information about any prior design family. Once trained, the network emits a geometry in **0.04 ms**, against the 48,000 stored simulations the standard pipeline requires beforehand.

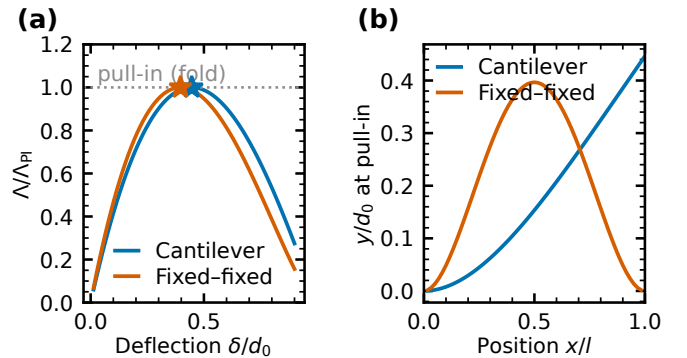
For device operation, a policy observing only a device's own tilt response infers the unmeasurable ceiling and recovers **12.8% more usable travel with 0.0% of devices destroyed**, over 1,536 held-out test episodes.

The same machinery addresses a question left open in the literature. Haluzan *et al.* (2010) conjectured that gains beyond simple gap shapes “are believed to be minimal” but had no means to test it. Free-form optimisation quantifies what remains: **0.14%** for the cantilever and **1.45%** for the fixed-fixed beam.

Evidence

The solver is validated against **eight independent sources published between 1967 and 2025**, of which two are the most discriminating. The exact gradient reproduces an analytic cube-scaling law to **0.00%**; this tests the differentiation itself rather than the forward solve, and would expose a plausible but incorrect adjoint. Second, the optimal gap exponent *reverses* between geometries, approximately 4/3 for a cantilever against exactly 1 for a fixed-fixed beam, and both are reproduced. Opposite trends in two geometries are not obtainable by fitting. Three of three instability mechanisms reported by Seeger and Boser are also predicted correctly.

The full implementation is released: solver, validation suite, RL environment, the MLP and Fourier-neural-operator baselines used for comparison, and a ninety-second demonstration notebook. No commercial licence is required at any stage, so the results reported here are reproducible on a laptop CPU.



The object we differentiate. The load maximum is pull-in, the saddle-node our solver targets directly.